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Universidade Politécnica de Macau
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應用科學學院
Faculdade de Ciências Aplicadas
Faculty of Applied Sciences

Doctor of Philosophy in Computer Applied Technology

Thesis Defence Examination

Thesis Review Comments

Student

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Title of Thesis			
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Reviewer

Name	Dr. Yapeng Wang	Date	2026-06-16
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Recommendation

☒ Accepted ☐ Suggestion for Revision*

* "Suggestion for Revision" will result in the delay of the oral defence.

Comments

Overview.

This thesis addresses the complexities of processing multi-turn, multi-speaker conversational data. The author systematically tackles issues like interleaved topics and fragmented meaning through innovative methods including Role-Aware Adapters (RAA), a hierarchical topic-driven framework (HiSum), a dynamic semantic optimization model (DABART), and a self-supervised critical model (SSCM). Additionally, it evaluates the practical limits of reasoning versus non-reasoning large language models (LLMs). Overall, it makes substantial contributions to natural language generation.

Knowledge Base and Literature Review.

The candidate demonstrates a profound understanding of natural language processing advancements. The literature review is exceptionally well-organized, categorizing research into generative paradigms, structural modeling, and reliability mechanisms. The author accurately traces the evolution of dialogue summarization from early architectures to

parameter-efficient fine-tuning and modern LLMs. Crucially, the candidate successfully identifies significant research gaps, specifically the shallow treatment of speaker roles, the difficulty of disentangling interleaved topics, and the challenge of factual hallucinations in high-stakes domains. This grounds the thesis in current academic debates.

Originality and Independence.

The originality of this thesis is highly commendable. The candidate engineers task-aligned structural solutions rather than simply applying larger models. The introduction of Role-Aware Adapters (RAA) models semantic interactions without overhauling underlying backbones. The DABART framework introduces highly original dynamic topic weighting to adapt to shifting semantic salience. To address reliability, the SSCM provides an innovative contrastive learning solution to filter out fluent but unsupported hallucinations. The critical assessment of reasoning-focused LLMs challenges prevailing assumptions, reflecting mature, independent, and forward-thinking scholarship.

Arguments and Reasoning.

The candidate presents highly robust and logical arguments. Each proposed method logically follows the identified limitations. For example, the argument that role-oriented summarization requires explicit semantic modeling directly motivates the RAA design. The premise that conversational topics are intertwined justifies the HiSum framework. The thesis employs a rigorous empirical methodology, systematically validating theoretical claims with comprehensive experiments across daily chat, medical, customer service, and financial datasets. The nuanced reasoning regarding why explicit LLM rationales often lead to verbosity and reduced factual grounding is thoroughly convincing.

Evaluation within the Field.

This thesis represents a highly significant contribution to dialogue-based Natural Language Generation. Dialogue summarization and hallucination mitigation are pressing challenges for deploying AI in high-stakes scenarios. The candidate's methodologies achieve highly competitive performance across established Chinese and English benchmarks. By proposing parameter-efficient solutions and self-supervised filtering mechanisms like SSCM, the research offers practical tools for industry applications where factual accuracy is paramount. The extensive LLM evaluation provides critical empirical guidelines regarding the limits of chain-of-thought prompting, confirming the high scientific merit of this work.

Completeness and Coherence.

The thesis is exceptionally coherent, characterized by a seamless narrative bridging theoretical conceptualization and empirical validation. The research objectives are systematically addressed as the candidate progresses naturally from static structural modeling for roles (RAA) and topics (HiSum) to dynamic semantic adaptation (DABART). It then effectively expands to encompass factual reliability in high-stakes environments using SSCM. Multi-metric automatic evaluations, rigorous human assessments, and deep ablation

studies guarantee the completeness of the empirical validation. The formatting and formulations are precise, leaving no unaddressed research questions.

Conclusion.

This is an high quality doctoral thesis tackling the complex problem of improving dialogue-based natural language generation. The candidate successfully developed innovative, structure-aware, and reliability-sensitive frameworks that significantly advance the capability of models to generate coherent and faithful summaries. The research is rooted in a thorough literature review, supported by rigorous experimental design, and offers tangible solutions for safe AI deployment in high-stakes domains. The work satisfies the requirements for a PhD degree, demonstrating the candidate's exceptional capacity for independent, high-impact scientific research.

Additional Comments.